Questions

Q1.

Figure 8 shows an atom of iron with its electron orbits.

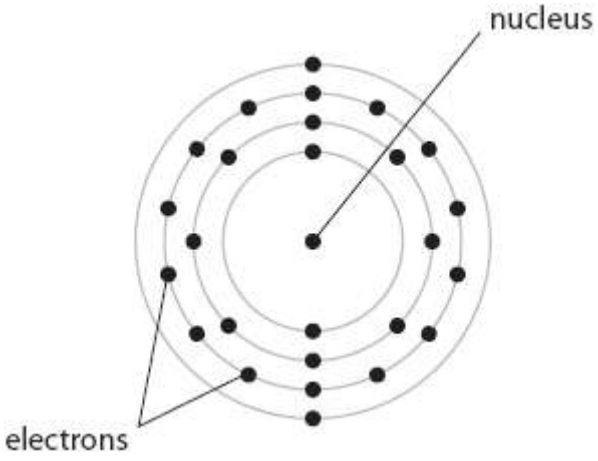

Figure 8

When iron is heated it glows red.

Explain what happens to the electrons during this process.

(3)

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(Total for question = 3 marks)

Q2.

An underground oil pipe starts to leak oil.
To find the leak, a technician adds a gamma source to the oil flowing in the pipe.

Describe how the technician can find the position of the leak.

(2)

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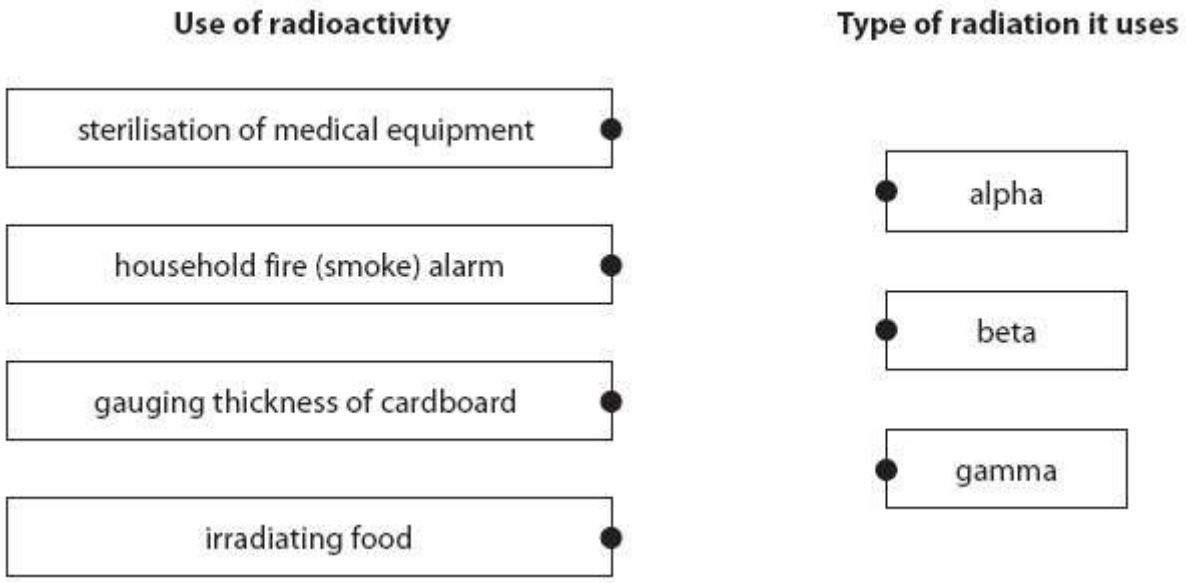
Q3.

Here are four uses of radioactivity.
Draw a line from each one of them to the type of radiation it uses.

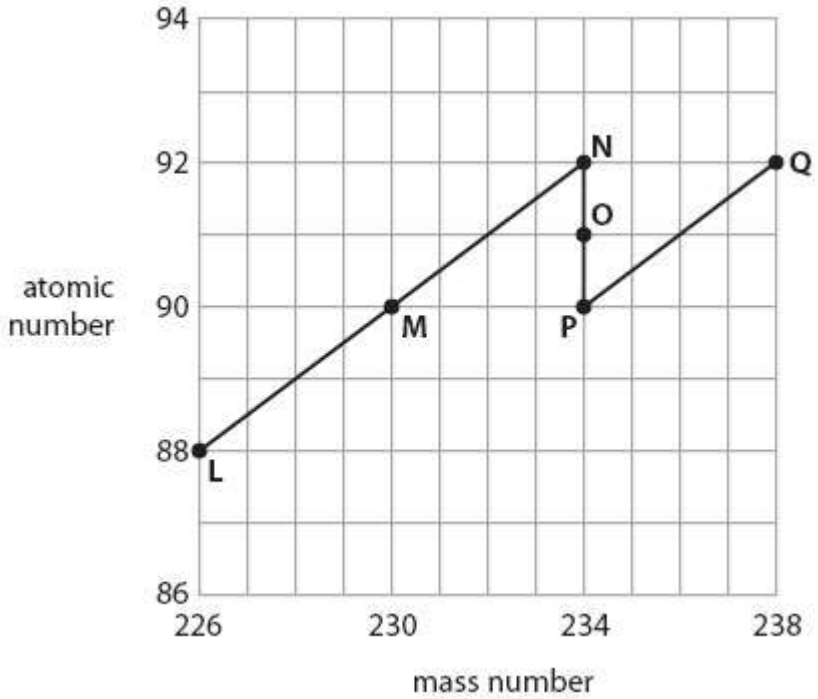
Each type of radiation may be chosen once, more than once or not at all.

(4)



Q4.

Uranium-238 is an isotope of uranium. It may undergo either radioactive decay or nuclear fission.
A nucleus of uranium-238 is shown as **Q** in the chart.



(a) State **two** letters from the chart which show isotopes of the same element.

(1)

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(b) Explain what happens when **Q** decays to **P.**

(2)

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(c) Explain what happens when **P** decays to **O**.

(2)

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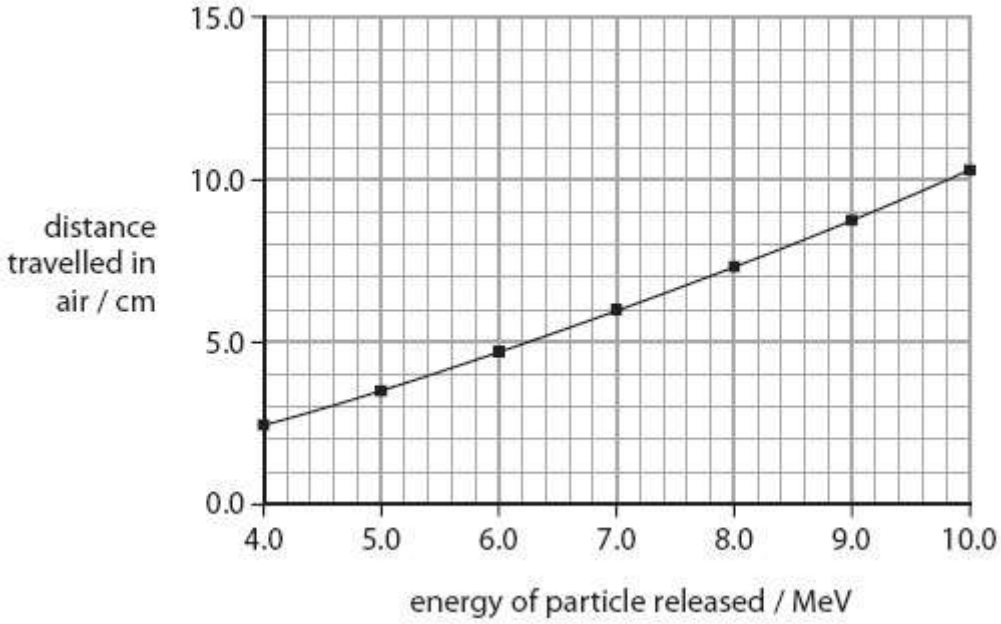
(d) Particles released during radioactive decay can have different energies.
A suitable unit for these energies is MeV.
For one type of decay, the particles released have energies between 4.0 MeV and 10.0 MeV.
The graph shows how far the particles with these energies travel in air.



(i) State the name of this type of particle.

(1)

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(ii) Use information from the graph to describe how the distance travelled in air depends on the energy of the particle.

(2)

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(e) Uranium-238 can only undergo nuclear fission by absorbing fast neutrons.
The fission emits neutrons which very quickly lose their energy.

Suggest why the fission of uranium-238 does not produce a chain reaction.

(2)

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(Total for Question = 10 marks)

Q5.

(i) Use words from the box to complete the sentences below about ions.

absorbing	gaining	inner	losing	outer

(2)

Atoms may form positive ions by ..... electrons.

The electrons involved in forming positive ions are the ..... electrons.

(ii) Which of these radiations is both electromagnetic and ionising?

(1)

☐ **A** alpha

☐ **B** beta minus

☐ **C** gamma

☐ **D** neutron

(iii) Which type of radiation will travel the shortest distance in air?

(1)

☐ **A** alpha

☐ **B** beta minus

☐ **C** beta plus

☐ **D** gamma

(Total for question = 4 marks)

Q6.

Lead-214 is a radioactive isotope.

(i) State **one** way in which radioactive isotopes can be harmful to people.

(1)

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(ii) Lead-214 emits β^- particles.

Describe what happens to the nucleus of a lead-214 atom when it emits a β^- particle.

(2)

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(Total for question = 3 marks)

Q7.

Every hospital radiographer who works with radiation wears a radiation badge.

The badge is used to monitor the amount of radiation the radiographer absorbs each month.

(i) Explain why it is important to monitor the amount of radiation a radiographer absorbs each month.

(2)

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(ii) Radiographers are restricted to a smaller annual dose of radiation nowadays compared to 50 years ago.

Complete the sentence by putting a cross (☒) in the box next to your answer.
This is because nowadays,

(1)

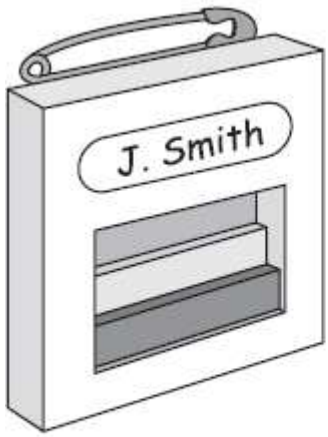
☐ **A** the radioactive sources have decayed

☐ **B** we can measure radiation more accurately

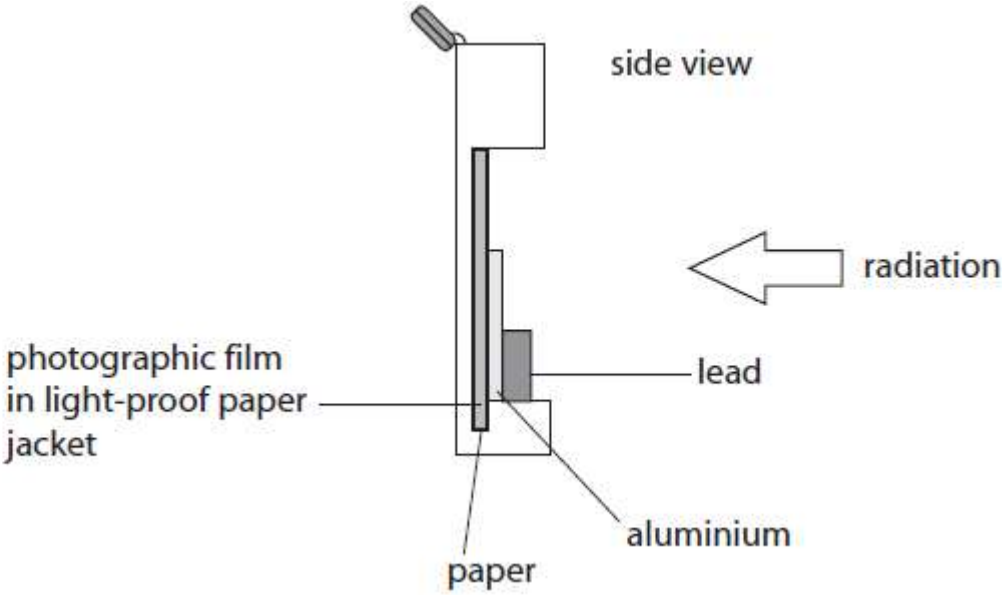
☐ **C** we have a better understanding of the risks from radiation

☐ **D** we have more effective ways of shielding against radiation

*(iii) The radiation badge contains a photographic film which is sensitive to radiation.



radiation badge



The radiation badge is sent to a laboratory after a month and the film is checked.
Explain how the badge shows the amount of different types of radiation that the radiographer has been exposed to.

(6)

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Q8.

Explain the use of a moderator in a nuclear reactor.

(3)

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(Total for question = 3 marks)

Q9.

The Sun is a star that is in a stable state called the main sequence.

The Sun, in its main sequence, radiates energy continuously but stays the same size.

(i) State the energy source in the Sun.

(1)

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(ii) Name the force that prevents the Sun from expanding.

(1)

.....

(Total for question = 2 marks)

Q10.

What is the approximate size of a hydrogen atom?

(1)

☐ **A** 10^{-3} m

☐ **B** 10^{-10} m

☐ **C** 10^{-19} m

☐ **D** 10^{-31} m

(Total for question = 1 mark)

Mark Scheme

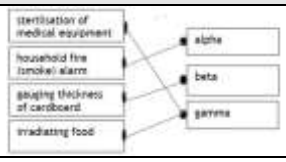
Q1.

Question number	Answer	Additional guidance	Mark
	An explanation that combines identification - application of knowledge (1 mark) and reasoning/justification - application of understanding (2 marks): • heat energy gained/absorbed by an electron (1) • electron changes orbit (1) • electron loses energy (1)	electron excited /unstable (by) emitting radiation	(3)

Q2.

	Answer	Acceptable answers	Mark
	A description to include • name of detector / move detector over the ground (1) • where leak is, there will be an increased rate (1)	(move) until a leak/high reading is found	**(2)**

Q3.

	Answer	Acceptable answers	Mark
		one mark for each correct line	**(4)**

Q4.

	Answer	Acceptable answers	Mark
(a)	P and M OR M and P OR N and Q OR Q and N	one mark for a pair	**(1)**
(b)	atomic /proton number drops by 2 **and** mass/nucleon	2 protons **and** 2 neutrons are lost 92 → 90 **and** 238	**(2)**

	number by 4 (1) (which is) alpha decay (1)	→ 234 helium nucleus given off (which is) alpha particle	
(c)	same mass/nucleon number but atomic/proton number increases by 1 (1) (negative) beta decay (1)	a neutron changes to a proton ignore GAINS a proton beta particle /electron given off	**(2)**
(d)(i)	alpha	Alpha ray, alpha particle, α Ignore capital letters	**(1)**
(d)(ii)	A description including two of one increases as other increases (1) rate of increase is in the range from 1.17 to 1.33 (cm/MeV) (1) range gradually increases more with energy (1)	the particles with higher energy travel further accept values quoted from graph not (quite) linear/not proportional /curves upwards accept values quoted from graph	**(2)**
(e)	chain reaction needs a neutron from one fission to reach another uranium nucleus/atom (at the right speed) (1) (fission of 238) needs fast/high(er) energy neutrons (1)	idea of continuous nature of chain reaction the neutrons would be going too slowly /do not have enough energy / lose energy too fast	**(2)**

Q5.

Question Number	Answer	Additional guidance	Mark
(i)	Atoms may form positive ions by **losing** electrons. (1) The electrons involved are the **outer** electrons (1)	accept any clear indication that correct word is in gap	**(2)**

Question Number	Answer	Mark
(ii)	The only correct answer is **C gamma** A is not correct because alpha radiation is not electromagnetic B is not correct because beta minus radiation is not electromagnetic D is not correct because neutron radiation is not electromagnetic	(1)

Question Number	Answer	Mark
(iii)	The only correct answer is **A alpha** B is not correct because beta minus travels further in air than alpha C is not correct because beta plus travels further in air alpha D is not correct because gamma travels further in air than than alpha and beta	(1)

Q6.

Question Number	Answer	Additional guidance	Mark
(i)	one from: (radiation from them) (can cause) cancer / tumours (1) radiation sickness / radiation poisoning (1) (radiation from them can) mutate / alter/ deform / damage / ionise / kill {cells OR DNA OR genes} (1) burns skin (1)	 accept any named type of cancer accept birth defects OR sterilisation Ignore unqualified poisoning kills you skin damage	(1)

Question Number	Answer	Additional guidance	Mark
(ii)	**neutron** (in the nucleus) (1) becomes a **proton** (and an electron) (1)	**down quark / d** (in the neutron) OR mass/nucleon number stays same becomes an **up quark / u** OR atomic/proton number increases by 1 n > p + e($^-$) scores 2 marks if no other mark scored allow for 1 mark (it) emits an electron OR beta (minus) is an electron OR energy is released OR loses a proton and gains a neutron IGNORE gaining/losing/becoming electron(s)	(2)

Q7.

Question Number	Answer	Acceptable answers	Mark
(i)	An explanation to include two from: Radiation is ionising (1) Radiation can cause specified damage e.g. cancer or damage/mutate DNA (1) if dose/exposure is too high (1)	(causes) ionisation/ (can) ionise/ mutate cells/tissue ignore radiation poisoning/death/make you ill ignore {damage/kill} cells/tissue if absorb(ing) too much (radiation) or so you don't absorb too much (radiation) Accept for both marks: Too much radiation can cause cancer (after a while)	**(2)**

Question Number	Answer	Acceptable answers	Mark
(ii)	☒ **C** we have a better understanding of the risks from radiation (1)		**(1)**

Question Number		Indicative Content	Mark
QWC	***(iii)**	An explanation including some of the following points • identification of alpha, beta, gamma as possible types of radiation • identification of X-rays as possible type of radiation • film is dark(er)/changes colour where radiation is absorbed • different areas of the film are exposed to different types of radiation • gamma (or X-rays) affect all areas of film • beta absorbed/stopped by aluminium/passes through paper • beta only reaches (top) part of film • alpha unlikely to be detected at all • the lead will stop (some of) gamma or (some) gamma will pass through lead/aluminium/paper • the paper will stop/absorb alpha throughout the question accept symbols for types of radiation	**(6)**

Level	0	No rewardable content
1	1 - 2	• a limited explanation which gives one relevant fact about types of radiation or the film badge e.g. types of radiation are alpha, beta and gamma OR beta absorbed by aluminium OR the radiation affects the film OR gamma can pass through lead • the answer communicates ideas using simple language and uses limited scientific terminology • spelling, punctuation and grammar are used with limited accuracy
2	3 - 4	• A simple explanation, giving more than one relevant fact about types of radiation OR the film badge OR at least one fact about both. e.g. The 3 types of radiation are alpha, beta and gamma. Gamma can pass through lead. OR The 3 types of radiation are alpha, beta and gamma. Radiation makes the film change colour. OR beta will get through the paper but alpha will be stopped (by paper). OR Radiation makes the film change colour. The lab. will compare how much got through the paper, aluminium and lead • the answer communicates ideas showing some evidence of clarity and organisation and uses scientific terminology appropriately • spelling, punctuation and grammar are used with some accuracy
3	5 - 6	• a detailed explanation giving more than two relevant points about the film badge OR at least one fact about the types of radiation AND more than one about the film badge e.g. Beta will get through the paper but alpha will be stopped (by paper).Gamma can penetrate the aluminium. OR The film detects radiation. The aluminium will stop beta but, not gamma. OR The 3 types of radiation are alpha, beta and gamma. Beta will get through the paper but alpha will be stopped (by paper). • the answer communicates ideas clearly and coherently uses a range of scientific terminology accurately • spelling, punctuation and grammar are used with few errors

Q8.

Question number	Answer	Additional guidance	Mark
	An explanation that combines understanding (1 mark) and reasoning/justification - understanding (2 marks): • slows down neutrons (1) • increase chance of collision / fission (1) • maintain the reaction rate (1)	 to allow capture by nucleus increase the reaction rate	**(3)**

Q9.

Question number	**Answer**	**Mark**
(i)	(nuclear) fusion	**(1)**

Question number	**Answer**	**Additional guidance**	**Mark**
(ii)	gravitational (pull)	gravity	**(1)**

Q10.

Question number	**Answer**	**Mark**
	B 10^{-10} m	**(1)**